Integration Medicine III: Coco and Oliver Student Reading

Integration Medicine III: Pre-Lecture Reading

Am I a Drama? Behavior is the presenting sign, not the diagnosis

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How to use this packet (plan 45 to 60 minutes): read Oliver first, then work Coco in order. Stop at every **Pause and think**. Write before you look anything up. We will work the answers in class. Do not copy a drug number from this packet as a standing order. Confirm every milligram-per-kilogram in Plumb or the hospital protocol.

Owner identity, phone numbers, emails, street addresses, clinician names, and clinic names have been removed. Dates are given as they appear on the laboratory and visit records. The records are incomplete and sometimes inconsistent, like the charts you will inherit. Noticing what is missing is part of the exercise.

How a doctor reads a chart

Keep a running problem list. After each stage write: what is new, how the differential changed, and the next test. Sort every sentence you write into one of four boxes:

Box	Meaning	Example that belongs here
Observed finding	A number, sign, or sentence that is actually in the record	Creatinine 0.3 mg/dL (Jan 2022)
Interpretation	What that finding might mean	Low creatinine can reflect small muscle mass or polyuria
Causal hypothesis	A proposed why that still needs a test	Cortisol excess is making the urine dilute
Diagnostic conclusion	A named disease with the test that proved it	A named UTI with a cystocentesis culture

A laboratory abnormality is not automatically the cause of the presenting complaint. A treatment response supports a hypothesis. It does not prove it. Two things that change in the

same week cannot be assigned one cause.

Therapeutic hierarchy for this course: identify and treat confirmed medical abnormalities while you relieve itch, pain, and sleep disruption. Reassess. Do not assume that a medical label or a behavioral label accounts for the whole dog.

Why these two cases

Owners rarely say “my dog has a thyroid problem” or “my dog’s calcium is off.” They say the dog freezes in the lobby, or paces all night and seems lost. Those are behavioral complaints. They are easy to label as training, anxiety, or old age.

The lecture idea is that **behavior is an integrative endpoint of multiple physiologic systems**. When a behavior is uncontrollable or refractory to environmental change and training, ask what is driving it from underneath before you treat the behavior itself.

Oliver is the case in which a building visit looks like “drama,” and the physiology is a three-step sympathetic storm. The same dog was hospitalized for parvovirus at 11 weeks. Coco is the case in which the owner sees night-time restlessness and “dementia,” and the chart underneath is a pile of endocrine, urinary, cardiopulmonary, and mineral problems. The first urinalysis looks like a urinary tract infection. The renal values stay normal. That combination is the trap.

Background to have in mind:

- The HPA axis mediates the endocrine stress response. A normal stress response is not the same as pathologic cortisol excess. Cortisol excess needs a clinical picture and a validated test.
- Thyroid hormone has broad metabolic and dermatologic effects. Links between canine hypothyroidism and a specific behavioral phenotype are less well established than the skin and coat signs.
- Calcium is tightly regulated. When calcium falls, PTH rises. Vitamin D status and dietary calcium-to-phosphorus ratio determine how well that system can compensate. Calcium also sets neuronal excitability.
- Itch is not only a skin event. IL-31 acts on sensory nerves. Whether that applies in a given dog has to be shown, not assumed.

Case 1. Oliver

Situational phobia, a three-step sympathetic storm, and a parvo hospitalization at 11 weeks

This section is built from the short Integration Medicine III Oliver deck (7 slides). The overview lecture also names a 2-year-old mixed-breed dog with panic confined to elevator use and a normal CBC, chemistry, and thyroid profile. The short deck does not reprint that laboratory screen. Write both sources down. Do not collapse them into one sentence until you decide they are the same visit.

Stage O1. The building sequence (printed as “Investigating Oliver’s situational phobia”)

The short deck walks the body through three places in one outing. Treat each line as an observed finding first, then as a proposed circuit.

Place	Observed finding on the slide	Circuit named on the slide	Interpretation named on the slide
Hallway	Panting, mydriasis, tachycardia, scanning the environment	Locus coeruleus–adrenal axis	Early sympathetic activation; heightened arousal; acute anxiety or catecholamine surge
Lobby	Refused to walk further, stood still; the owner carried him	Periaqueductal gray (PAG) and amygdalo-cortical circuits	Behavioral freezing; motor inhibition from fear paralysis
Concierge	Hyperthermia, severe tachycardia, sweaty paw pads	Sympathoadrenal hyperactivation	Catecholamine surge with a thermogenic and cardiovascular response

This is still a stress response until a validated adrenal test says otherwise. Hyperthermia and sweaty pads at the concierge are not a fever workup by themselves, and they are not proof of heatstroke. They are the same axis turning up.

Pause and think

1. Write a one-sentence problem representation that names the three places and the new sign that appears at each step.

2. Sort each row of the table into the four boxes (observed finding, interpretation, causal hypothesis, diagnostic conclusion). Which column is already a conclusion you have not proved?
3. What would make this no longer “situational only” tonight? Name history and exam findings, not a drug.
4. If the overview slide says elevator and this deck says hallway, lobby, and concierge, what do you write in the problem list?
5. Design a desensitization plan that starts in the hallway and does not start with a daily SSRI.

Stage O2. The 11-week parvo record (4 Jan 2021)

The short deck then shows a medical record from when Oliver was 11 weeks old. Diarrhea and vomiting had stopped. He was lethargic and inappetent. The owner was concerned about progression.

Parameter	Printed value
Weight	4.15 lb
Temperature	101 F
Mucous membranes	Pale, tacky
CRT	2 to 3 seconds
Heart	120 / min, regular, strong synchronous femoral pulses, no murmur or arrhythmia
Lungs	Normal bronchovesicular sounds
Abdomen	Soft, nonpainful, no masses
Hydration	5 to 10 percent dehydrated
Neurologic	Alert, however generalized weakness
Body condition	3.5 / 9
Coat	Unkempt, no ectoparasites
Urogenital	Male intact

Assessment as written: parvovirus, dehydrated, lethargic, generalized weakness.

Plan as written: the clinic could offer fluids, blood work, day hospitalization, antimicrobials, anti-nausea medication, a stomach protectant, an appetite stimulant, and a blood-pressure check. The estimate was over $1,000 and was described as not gold-standard for a sick parvo puppy because of limited capacity. A tertiary-hospital estimate of about $6,000 to $7,000 was not affordable. The owner agreed to take him to a lower-cost referral hospital.

Pause and think

1. Which problems are urgent on this page, and which can wait?
2. Why does “no longer having diarrhea” not take parvovirus off the list?
3. Pale tacky membranes, CRT 2 to 3 seconds, and 5 to 10 percent dehydration: what do you do first, and what number do you not invent?
4. How does a week in a hospital at 11 weeks change the socialization window of 3 to 12 weeks?
5. Write the two sentences you would say to the owner about gold-standard care versus the care you can actually deliver tonight.

Stage O3. After parvo (25 Jan 2021)

Twenty-one days later the record says the historical parvo diagnosis is cleared.

Parameter	Printed value
Weight	5.8 lb
Temperature	101.5 F
Mentation	Bright, alert, responsive; hydrated
Mucous membranes	Pink, moist; CRT 1.5 seconds
Heart	100 / min, regular sinus rhythm, no murmur, pulses strong and synchronous
Lungs	Normal, no increased effort
Neurologic	Normal mentation, no nystagmus or head tilt, cranial nerves intact
Musculoskeletal	Ambulatory in all four limbs, normal gait, BCS 4 / 9
Coat	Healthy, no ectoparasites

Plan as written: DA2PP plus lepto in the right forelimb in three weeks; oral Bordetella; a topical parasite preventive, with a later conversation about an oral chewable; consider Lyme later

because the family has a house upstate; return in three weeks for the DA2PP plus lepto booster and rabies.

Pause and think

1. What did 21 days change on the exam, and what did it not prove about future fear?
2. A recovered parvo puppy is not a behavior diagnosis. Which follow-up questions protect the 3-to-12-week window now?
3. The overview slide lists a later normal CBC, chemistry, and thyroid. Which of those, if any, belongs on 25 Jan 2021?

Stage O4. Problem list and differentials (as printed)

The short deck lists, on one slide, the problems, differentials, and tests to rule out:

- Poor socialization during the critical window of 3 to 12 weeks
- Generalized anxiety disorder
- Thunder phobia
- Noise aversion
- Situational phobia
- Trauma
- Hospitalized for parvovirus infection at 11 weeks old

That slide does not print the tests. It also does not print a single winner. Situational phobia is one row, not the diagnosis until you have watched the context and excluded the others.

Pause and think

1. Which items on that list are observed history, and which are diagnoses that still need criteria?
2. Can situational phobia and thunder phobia both be true? What history separates them?
3. What medical tests would you still run before you keep a behavior-only plan, and why?
4. If the dog freezes only in this building, which rows become less likely, and which stay open?

Stage O5. Pharmacology recap (class matching, not a prescription)

The last slide is a simplified matching chart. It names drug **classes**, not milligram-per-kilogram standing orders. Confirm every number in Plumb or the hospital protocol.

Behavior category	Brain region focus	Main transmitter to modulate	Drug class on the slide
Panic / noise phobia	Locus coeruleus, PAG	Decrease norepinephrine	Alpha-2 agonist
Chronic anxiety	Amygdala, cortex	Increase serotonin	SSRI / TCA
Fear aggression	Amygdala, PAG	Increase serotonin and GABA	SSRI plus a benzodiazepine
Compulsive disorder	Basal ganglia, cortex	Increase serotonin, decrease dopamine	SSRI, clomipramine
Cognitive dysfunction	Hippocampus, cortex	Increase dopamine and acetylcholine	Selegiline
Social fear / trauma	Amygdala, locus coeruleus	Increase serotonin, decrease norepinephrine	SSRI plus clonidine

Pause and think

1. Using only the building-sequence signs, which row of this table do you start with, and which row do you not start with?
2. Why is a daily SSRI the wrong first move if the panic is still one lobby?
3. Clonidine appears in two rows (alpha-2 agonist; social fear / trauma). What history would make you consider it, and what would you check first?
4. Coco later carries a cognitive-dysfunction label. Which row of this table is about Coco, and why is it dangerous to give Oliver that row?

Teaching point: The hallway is early sympathetic activation. The lobby is freezing. The concierge is a catecholamine surge you can see on the pads. A parvo hospitalization at 11 weeks is history that belongs on the problem list. It is not, by itself, the diagnosis of the freeze. Drug class follows the circuit you have actually localized.

Management principles (not a recipe):

- Localize the step (hallway versus lobby versus concierge) before you name a drug class.
- Desensitization and counterconditioning start at the earliest step the dog can still eat.

- Situational anxiolysis or a serotonergic trial is a hospital-protocol conversation.
- Reassess after the context is trained. If the dog now panics everywhere, reopen the medical screen and the differential on the problem-list slide.

Case 2. Coco

Cushing's, diabetes insipidus, a UTI, "dementia," and what lies underneath

Coco was a real patient. The records come from an integrative primary-care practice, a second primary-care site, an emergency and specialty hospital, an outside radiology service, and outside laboratories (Antech, IDEXX, Nelco). They do not agree with each other on every line. That is the point.

Stage 0. Signalment and the history you wish you had

Patient: Beagle, female spayed. Date of birth 4 Nov 2011. She enters this packet at 10 years 2 months (Jan 2022) and leaves it at 11 years 9 months (15 to 17 Aug 2023).

Coat: red and white (later primary-care face sheet).

Microchip: none recorded on the emergency face sheet.

Weight history, printed:

Date	Weight
Apr 2021	about 39.6 lb (pre-lecture compilation)
Jun 2021	34.2 lb
Jan 2022	32.8 lb
22 Jun 2022	29.6 lb
29 Jun 2022	29.8 lb
23 Sep 2022	29.0 lb
Jan 2023	30.4 lb
9 Feb 2023	30.2 lb
13 Mar 2023	28.8 lb
27 Mar 2023	25.8 lb
17 to 18 Apr 2023	23.8 lb (lowest printed)

Date	Weight
19 Jun 2023	29.6 lb
15 Aug 2023 check-in	14.1 kg on the visit banner; 14.8 kg on the vitals line

The record does not explain the roughly 14 percent fall in spring 2021. The 22 percent fall from January to mid-April 2023, then the rebound by June, is in the later face sheet. The August emergency banner and the vitals line do not match. Write both down. Do not pick the one that is convenient.

Vaccine titers (Antech, received 27 Jan 2022, reported 28 Jan 2022): distemper vaccine titer positive; parvovirus vaccine titer positive. The laboratory comment: appropriate immunologic response (titer equivalents CDV at least 1:16 by SN; CPV at least 1:80 by HA).

Allergy panel (Nelco, received 14 Sep 2021, reported 15 Sep 2021): environmental panel with high-positive and low-positive reactions among grasses, weeds, trees, molds, and danders (OCR of the fax is incomplete on several allergen names). Food panel: eggs high-positive. The approved-food list printed by the laboratory names beef, chicken, eggs, corn, peanut, and carrots as LP or HP. She later received Cytopoint. A serum allergy panel is not a diet trial and is not a diagnosis of cutaneous adverse food reaction.

Diagnoses later typed onto a June 2023 face sheet (order as printed):

Date entered	Label as printed
9 Jun 2021	Phimosis (in a spayed female; notice it)
27 Apr 2022	Hyperadrenocorticism
18 May 2022 and 26 May 2022	Diabetes insipidus (entered twice)
8 Feb 2023	Hepatic disease
17 Mar 2023	Hematuria
16 Apr 2023	Cognitive dysfunction or dementia
23 Apr 2023	Hyperthyroidism
29 Apr 2023	Lipoma
27 May 2023	Hyperadrenocorticism (again)

The tests behind several of those labels are not in the records provided.

Pause and think

What history would you take before ordering anything? Write the questions, not the tests. Include diet and treats, water intake in measured cups or milliliters over 24 hours, how often and how much she urinates, whether she is up at night and why, appetite, panting, cough, exercise tolerance, skin and coat, current drugs and supplements, and the order in which each sign appeared.

Stage 1. First laboratory data (Jan 2022)

Antech Vet Screen, received 27 Jan 2022. The laboratory noted **4+ hemolysis and 4+ lipemia** and reported **no significant interference**. No urine specific gravity is recorded with this chemistry.

Test	Result	Reference interval	Flag
Total protein	5.8 g/dL	5.0 to 7.4	
Albumin	3.8 g/dL	2.7 to 4.4	
Globulin	2.0 g/dL	1.6 to 3.6	
A/G ratio	1.9	0.8 to 2.0	
AST	13 IU/L	15 to 66	Low
ALT	28 IU/L	12 to 118	
Alkaline phosphatase	243 IU/L	5 to 131	High
Total bilirubin	0.1 mg/dL	0.1 to 0.3	
BUN	14 mg/dL	6 to 31	
Creatinine	0.3 mg/dL	0.5 to 1.6	Low
BUN:creatinine ratio	47	4 to 27	High
Phosphorus	5.4 mg/dL	2.5 to 6.0	
Glucose	82 mg/dL	70 to 138	
Calcium	9.3 mg/dL	8.9 to 11.4	
Sodium	148 mEq/L	139 to 154	
Potassium	4.6 mEq/L	3.6 to 5.5	
Na/K ratio	32	27 to 38	
Chloride	108 mEq/L	102 to 120	
Cholesterol	317 mg/dL	92 to 324	Top of interval

Test	Result	Reference interval	Flag
CPK	86 IU/L	59 to 895	
LDH	259 IU/L	50 to 500	

Urine protein:creatinine (same accession): protein 30 mg/dL, creatinine 79.4 mg/dL, ratio **0.4** (laboratory cutoff printed as below 0.5). The interpretation block says a UPC must be read with sediment; hematuria or inflammation can raise it; ratios at or above 0.5, repeated, and not pre-renal or post-renal, are overt proteinuria; ratios at or above 2.0 in dogs are usually glomerular.

A second UPC on 31 Mar 2022 (Antech): protein 30.6 mg/dL, creatinine 80 mg/dL, ratio **0.4**.

Pause and think

1. Which abnormalities matter, and which are probably noise?
2. Creatinine is 0.3 mg/dL. What can cause a low creatinine, and how does it distort the BUN:creatinine ratio?
3. Give a differential for the ALP. What else on the panel supports or argues against each item?
4. Cholesterol at the top of the interval plus a 4+ lipemic sample: what do you want to know about this blood draw?
5. Which analytes can hemolysis or lipemia distort, even when the laboratory reports no significant interference?
6. Which urine tests do you still need, and how should the sample be collected?

Stage 2. The polyuria workup (Apr to Jun 2022)

The problem list now contains hyperadrenocorticism (entered 27 Apr 2022) and diabetes insipidus (entered 18 May and again 26 May 2022). The tests behind those labels (ACTH stimulation, low-dose dexamethasone suppression, urine cortisol:creatinine, imaging, desmopressin trial) are **not in the records provided**. In June 2022 the owner reported that Coco was drinking about **eight cups of water a day**.

Differential for polyuria and polydipsia	What would support it	What you would run
Hyperadrenocorticism	High ALP, high cholesterol, panting, dilute urine, stress leukogram	ACTH stimulation or dexamethasone suppression; urine cortisol:creatinine; abdominal ultrasound
Central diabetes insipidus	Persistently dilute urine, normal renal values, response to desmopressin	Desmopressin trial; modified water deprivation only when it is safe
Nephrogenic diabetes insipidus	Dilute urine that does not respond to desmopressin; a systemic cause such as hypercalcemia, hypokalemia, pyelonephritis, hepatic disease	Calcium, potassium, culture, imaging, desmopressin trial
Chronic kidney disease	Azotemia, isosthenuria, hyperphosphatemia	Creatinine, SDMA, urinalysis, blood pressure, imaging
Urinary tract infection or pyelonephritis	Pyuria, bacteriuria, frequent urination, systemic signs	Cystocentesis urinalysis with sediment and culture
Diabetes mellitus	Hyperglycemia and glucosuria	Glucose, urinalysis
Hypercalcemia	High calcium, azotemia later	Total and ionized calcium, PTH
Hepatic insufficiency	Low BUN, low albumin, high bile acids	Bile acids, ammonia, ultrasound
Primary polydipsia	Normal renal and endocrine tests	Diagnosis of exclusion
Drugs	Glucocorticoids, diuretics	Complete drug history

Pause and think

1. Eight cups is about 1.9 L. At roughly 13.5 to 14 kg, what is that in mL/kg/day, and how does it compare with the usual threshold for polydipsia?
2. Define hyposthenuria, isosthenuria, and inadequately concentrated urine in a dog. Which would you expect in each row of the table?
3. Which two or three tests give you the most information first, and why?
4. How can Cushing’s and diabetes insipidus coexist or mimic each other?
5. What do you do when a diagnosis is recorded in the chart but the test behind it is not?

Stage 3. The urinalysis said UTI, but the kidneys were fine (Jun 2022)

This is the stage the lecture wants you to be able to teach back.

What the primary-care chart actually says, in order:

- **2 Jun 2022:** cystocentesis. Urine culture with urinalysis submitted to Antech.
- **6 Jun 2022:** the clinician reports that the laboratory isolated a hemolytic *Streptococcus*. The owner is told Coco is a little better on an increased dose of "meds." Sensitivity is still pending. Enalapril 5 mg twice daily is dispensed the same day. The indication is not documented.
- **7 Jun 2022:** a second telephone note says staphylococcal UTI, sensitive to cefpodoxime. A 14-day course of cefpodoxime 100 mg once daily is started. The two organism names do not agree.
- **9 Jun 2022:** the owner reports that Coco is not finishing breakfast and walked away from dinner. The clinician suspects the antibiotic.
- **13 Jun 2022:** Coco is lethargic. The owner is told to stop **all medication**, including the trilostane (Vetoryl) that treated her Cushing's.
- **14 Jun 2022:** three blood-pressure readings: 87/50, 105/76, and 101/75 mmHg.
- **22 Jun 2022:** urine specific gravity **1.008**. Desmopressin ophthalmic drops started that day. Another culture-if submitted. Weight 29.6 lb.
- **24 Jun 2022:** repeat urinalysis reported with **no bacteria**. The owner says she is doing much better on the desmopressin drops and that PU/PD is less.

Renal values across the whole record stay in the non-azotemic range until the last illness: creatinine 0.3 mg/dL (Jan 2022), 0.4 to 0.6 mg/dL (Jun 2023 IDEXX comparisons), 0.8 mg/dL (15 Aug 2023). BUN 14 (Jan 2022), 21 and 21 (13 and 22 Jun 2023), 41 (15 Aug 2023, first high BUN). Phosphorus is not high in 2022.

Pause and think

1. How does a dog with a bacterial UTI have a normal creatinine and BUN? What does a normal creatinine fail to exclude?
2. Why is cystocentesis the preferred sample, and what would a free-catch sample change?
3. Dilute urine and glucocorticoid excess both predispose to UTI. Explain why. Does that change how long you treat?
4. Distinguish bacterial cystitis, pyelonephritis, subclinical bacteriuria, and sterile cystitis. Which would you treat, and on what evidence?

5. Lethargy, inappetence, and low-normal blood pressure appeared within weeks of several new drugs. List every plausible cause and say how you would separate them.
6. A UTI is treated and her renal values are normal. Is the polyuria explained? Why or why not?
7. The chart names two different organisms one day apart. What do you write in the problem list tonight: “UTI,” “possible UTI,” or “conflicting culture reports”?

What a doctor does not do with this stage: treat the UTI, see the creatinine stay “normal,” and close the polyuria. Lower urinary infection can sit on top of dilute urine. It does not create eight cups of water a day by itself unless you have also proved pyelonephritis or another concentrating defect.

Stage 4. The desmopressin trial and confounding (Jun 2022 to Jan 2023)

On 22 Jun 2022 the USG was 1.008 and desmopressin ophthalmic drops were started. By 29 Jun the owner reported that daily intake had fallen from eight cups to two. The USG that day was **1.023**. The same visit notes that she was now **off trilostane and all other medication**, and that the last two nights she had panting, shallow fast breathing, and was “a little out of it,” with no vomiting, diarrhea, coughing, sneezing, or (by that note) PU/PD.

Printed in-house USG values:

Date	USG	Context in the chart
22 Jun 2022	1.008	Desmopressin started; UTI antibiotic just finished
29 Jun 2022	1.023	One week of drops; 8 cups to 2 cups; off trilostane and other drugs
7 Jul 2022	1.009	Single sample, no comment recorded in the pre-lecture compilation
5 Aug 2022	1.015	pH 6.5; cranberry discussed
19 Aug 2022	1.020	pH 7
26 Aug 2022	1.019	pH 6.5
23 Sep 2022	1.007	Weight 29.0 lb

Date	USG	Context in the chart
30 Sep 2022	1.027	“Drops are working”; UPC 0.4; continue desmopressin; pituitary laser recorded
Oct to Nov 2022	1.022 to 1.026	Stable on desmopressin (pre-lecture compilation)
1 Dec 2022	1.020	
8 Dec 2022	1.026	
21 Dec 2022	1.011	Leukocytes; cefpodoxime started for UTI
28 Dec 2022	1.017	
4 Jan 2023	1.008	Urinating constantly; desmopressin raised from 1 drop three times daily to 2 drops twice daily
6 Jan 2023	1.010	Walked every two hours the night before
25 Jan 2023	1.006	Owner: peeing all day and all hours; a possible sterile cystitis had been noted in December

A September 2022 note also records ALP still high at **875 IU/L**, urine protein 2+, UPC later 0.7, and enalapril restarted at 5 mg once daily. That UPC is not the 0.4 from January and March.

Pause and think

1. In June 2022 three things changed within about two weeks: trilostane was stopped, a UTI was treated, and desmopressin was started. What can you conclude about the cause of her improvement, and what can you not conclude?
2. Does a good response to desmopressin prove central diabetes insipidus? What else can respond?
3. In December 2022 to January 2023 the USG fell again despite a higher desmopressin dose. List at least six reasons the response could fail.
4. “Peeing all the time” can mean polyuria or pollakiuria. What history and what test separate them?
5. “Laser of the pituitary” is in the chart. What does that sentence do to your problem list, and what does it not prove?

Stage 5. What the owner sees at night

Printed behavioral and sleep observations:

- Jun 2022: panting, shallow fast breathing, and “a little out of it” on two consecutive nights (the same week she came off trilostane).
- Jan 2023: taken out every two hours overnight; waking in the middle of the night.
- 16 Apr 2023: cognitive dysfunction or dementia is entered on the face sheet. No screening scale is in the records provided.
- By Jun 2023: sedatives and CBD had been used for sleep and had been stopped; “still up at night.”
- 15 Aug 2023 emergency history: abnormal mentation, intermittent stiffness, legs giving out, twitching. The owner did **not** report increased thirst or urination that week.

Possible contributor	Evidence in the chart	How you would test or treat
Nocturia from DI, UTI, or cystitis	USG 1.006 to 1.011 with leukocytes; up every two hours	Repeat sediment and culture; measure intake and output; check desmopressin adherence
Hyperadrenocorticism	Panting, high ALP, off trilostane from Jun 2022	Re-establish the diagnosis and monitor therapy
Pain	Degenerative joint disease of the knee, elbow, and spine; tramadol, Adequan, laser	Orthopedic exam, analgesic trial
Cardiopulmonary disease	Grade IV murmur later; heart disease; bronchitis; possible tracheal collapse	Sleeping respiratory rate, echocardiogram, radiographs
Medications	Furosemide as needed, a prednisolone-containing bronchitis drug, theophylline, desmopressin, selegiline	Medication reconciliation
Sensory decline	Lenticular opacities in both eyes	Ophthalmic and hearing assessment
Mineral and thyroid disturbances	Calcium, phosphorus, vitamin D3, thyroxine on and off the drug list	Ionized calcium, PTH, phosphorus, 25-hydroxyvitamin D, thyroid panel
Canine cognitive dysfunction	Diagnosis entered Apr 2023; no documented screening scale	Validated questionnaire after treatable causes are addressed

Possible contributor	Evidence in the chart	How you would test or treat
Intracranial lesion	The internist raised a pituitary tumor in Aug 2023; abdominal ultrasound found no adrenal mass	Advanced imaging, neurology referral

Pause and think

1. Which of these can you rule in or out from the chart alone?
2. “Dementia” was recorded while at least four treatable contributors were still active. What does the therapeutic hierarchy say about order?
3. What would you tell the owner on the first night-time visit, in two sentences that do not promise a diagnosis?

Stage 6. Weight loss and an accumulating problem list (Jan to Jun 2023)

See the weight table in Stage 0. Liver and lipid values over time (reference intervals **differ between laboratories**):

Date	ALP (IU/L)	ALT (IU/L)	GGT (IU/L)	Cholesterol (mg/dL)	Laboratory
Jan 2022	243 (RI 5 to 131)	28 (12 to 118)	not run	317 (92 to 324)	Antech
Sep 2022	875	not recorded	not recorded	not recorded	Primary-care note
13 Jun 2023	1,177 (IDEXX comparison column)	226	16	294	IDEXX comparison
22 Jun 2023	1,193	169	19	321	IDEXX comparison
15 Aug 2023	334 (RI 23 to 212)	93 (10 to 125)	5 (0 to 11)	215 (110 to 320)	IDEXX Catalyst

Bilirubin stayed within the printed interval on these dates (0.1, 0.5, 0.1, 0.4 mg/dL).

June 2023 medication list as the owner stated it: pimobendan twice daily, furosemide as needed, desmopressin 2 drops twice daily, a hepatic support supplement, and vitamin D3 4 drops daily. She had stopped selegiline, thyroxine, calcium lactate, sedatives, and CBD.

Pause and think

1. Give a differential for 22 percent weight loss in three months. What would you examine and test first?
2. ALP above 1,100 with a GGT of 16 to 19 and a normal bilirubin: what does that pattern suggest, and what would you do to confirm it?
3. Hyperthyroidism is rare in dogs, yet a thyroxine supplement appears on the drug list and “hyperthyroidism” is typed on the face sheet. How do you reconcile those two lines?
4. For each drug on the June 2023 list, write the indication, a likely effect on urine concentration, and a risk.
5. Why might three drugs stopped together make the next month uninterpretable?

Stage 7. Calcium, phosphorus, vitamin D, and the parathyroid

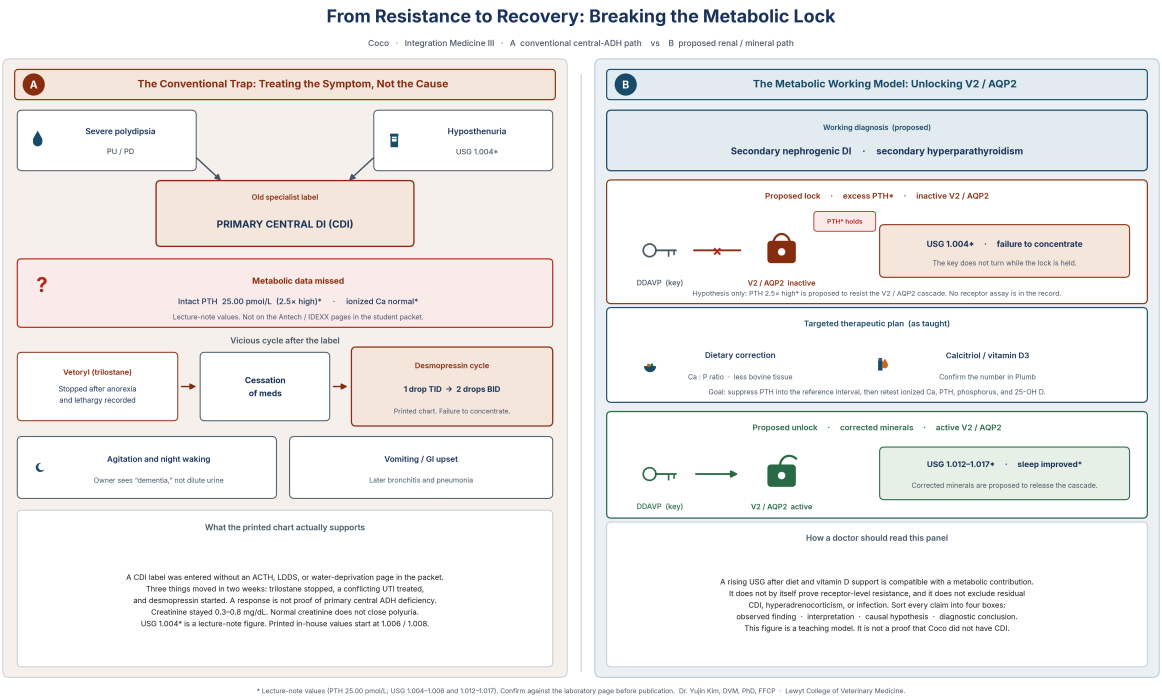
Printed mineral and renal pairs:

Date	Calcium (mg/dL)	Phosphorus (mg/dL)	Creatinine (mg/dL)	BUN (mg/dL)	Calcium × phosphorus
Jan 2022	9.3 (RI 8.9 to 11.4)	5.4	0.3	14	50
13 Jun 2023	10.3 (RI 7.9 to 12.0)	5.4	0.6	21	56
22 Jun 2023	9.5	6.4	0.4	21	61
15 Aug 2023	10.8	7.3 (high; RI 2.5 to 6.8)	0.8	41 (high; RI 7 to 27)	79

Calcium lactate and vitamin D3 drops were on the 2023 medication list. When they were started, why, and at what dose are not recorded. **Parathyroid hormone, ionized calcium, 25-hydroxyvitamin D, and a diet analysis are not in the laboratory PDFs provided with this packet.**

The lecture slides list values that are **not on those PDFs**: PTH 25 pmol/L (slide reference 1.1 to 10.6), total calcium 8.8 mg/dL, ionized calcium 1.30 mmol/L, and a USG change 1.006 to 1.017 after vitamin D3. Treat those as lecture-note figures to be reconciled in class, not as findings you have already verified. The August 2023 IDEXX total calcium is 10.8 mg/dL, not 8.8. That is a chart conflict. Write it down.

Figure 1. Conventional CDI path versus the metabolic working model



From Resistance to Recovery: Breaking the Metabolic Lock

Two-panel teaching figure. Left: the conventional specialist path that labeled primary central diabetes insipidus, then stopped trilostane and raised desmopressin from 1 drop three times daily to 2 drops twice daily (printed chart). Right: the lecture working model of secondary nephrogenic DI and secondary hyperparathyroidism, with a proposed V2 / AQP2 lock. Values marked * are lecture-note figures (PTH 25.00 pmol/L; USG 1.004–1.006 and 1.012–1.017). They are not on the Antech / IDEXX pages in this packet. The V2 / AQP2 lock is a hypothesis, not a receptor assay. A rising USG after diet and vitamin D support does not prove receptor-level resistance and does not exclude residual CDI, hyperadrenocorticism, or infection. Confirm every milligram-per-kilogram in Plumb. Do not copy a calcitriol number from a lecture slide as a standing order.

Pause and think

1. Before you supplement calcium and vitamin D in a dog, which measurements must you have, and why?
2. Total calcium is a poor surrogate for the biologically active fraction. Which result would you request instead?
3. Her creatinine is “normal” until the last day. How would you separate renal, nutritional, and primary causes of an elevated PTH? Is a normal creatinine enough to exclude a renal cause?
4. How might chronic glucocorticoid excess change intestinal calcium absorption, urinary calcium loss, and PTH?

5. Phosphorus climbs from 5.4 to 7.3 mg/dL while she is on vitamin D3. Give two explanations and a plan.
6. A product of calcium and phosphorus near 80 is often cited as a risk for soft-tissue mineralization. What are the limits of that rule, and what else is happening on 15 Aug (hydration, BUN, respiratory disease)?
7. What would you ask about her diet, and why does “raw diet with bone” not equal a diagnosis?

Stage 8. The final illness (Jun to Aug 2023)

In June 2023 Coco vomited and aspirated, developed pneumonia with a productive cough, and was hospitalized for three days (history taken at the August emergency visit). Heart disease was described as worsening. She was on theophylline and had received a carbapenem during that hospitalization.

15 Aug 2023, emergency service, 11 years 9 months. Chief complaint: a few days of increased respiratory rate and effort and noted stridor. Legs giving out. Twitching at home. Mentation abnormal, “out of it,” intermittent stiffness. Vomited once. Less interested in food. No diarrhea or sneezing. **Owner reported no PU/PD that week.** History given to the emergency clinician: bronchitis, recently prescribed temaril-P and theophylline; recent pneumonia treatment; history of diabetes insipidus, hyperadrenocorticism, and UTIs.

Vitals (printed):

Parameter	Value
Weight	14.8 kg on the vitals line; 14.1 kg on the visit banner
Temperature	101.9 F
Heart rate	190 / min
Respiratory rate	40 / min
Respiratory effort	Abnormal
Mucous membranes	Pale and tacky
Capillary refill	2 to 3 seconds
Blood pressure	130 (no units/site recorded on the exam sheet)

Exam (printed): about 5 percent dehydration; lenticular opacities both eyes; grade IV heart murmur; moderate stridor and increased bronchovesicular sounds both sides; mild respiratory effort; abdomen soft and nonpainful; ambulatory in all four limbs with intact conscious

proprioception and normal cranial nerves; quiet, alert, responsive. Oral exam limited by stress. Rectal examination not performed. Animal Trauma Triage and Modified Glasgow Coma Scale fields are blank.

ECG (first note): ventricular tachycardia with intermittent premature complexes. Later internal-medicine notes call the collapse rhythm **supraventricular tachycardia**. Write both. Do not collapse them into one word.

Thoracic radiographs (DACVR STAT report, 15 Aug 2023): four images. Mild to moderate broncho-interstitial pattern; mild unstructured interstitial pattern in the right middle lung lobe; cardiac silhouette and pulmonary vasculature unremarkable; some variation in tracheal height; cranial mediastinum widened on VD/DV without tracheal deviation, suspected fat; stomach contains heterogeneous soft tissue and granular mineral; elbow degenerative change; incidental ventral spondylosis. Conclusions: (1) bronchitis (irritant, infectious, or allergic) with bronchopneumonia; (2) possible tracheitis or atypical dynamic tracheal collapse; (3) gastric contents, food and/or foreign material; (4) elbow degenerative joint disease. The radiology header lists breed as “bulldog” and a birthdate of 15 Aug 2012. The emergency face sheet says Beagle, born 4 Nov 2011. That is a record error. Notice it.

IDEXX ProCyte and Catalyst, 15 Aug 2023, with the analyzer’s own June 2023 comparison columns. Selected printed values:

Test	15 Aug 2023	Reference interval	22 Jun 2023	13 Jun 2023
RBC	10.00 M/µL	5.65 to 8.87	8.47	8.87
Hematocrit	56.7 %	37.3 to 61.7	57.1	55.5
Hemoglobin	21.0 g/dL	13.1 to 20.5	19.5	20.6
MCV	56.7 fL	61.6 to 73.5	67	62.6
RDW	23.2 %	13.6 to 21.7	21.0	(not printed)
Reticulocytes	20.0 K/µL	10.0 to 110.0	51	38.1
WBC	13.96 K/µL	5.05 to 16.76	9.6	10.75
Neutrophils	12.14 K/µL	2.95 to 11.64	7.699	9.14
Lymphocytes	1.10 K/µL	1.05 to 5.10	1.421	0.93
Eosinophils	0.02 K/µL	0.06 to 1.23	0.029	0.00
Platelets	491 K/µL	148 to 484	432	325
Glucose	111 mg/dL	70 to 143	98	114
Creatinine	0.8 mg/dL	0.5 to 1.8	0.4	0.6

Test	15 Aug 2023	Reference interval	22 Jun 2023	13 Jun 2023
BUN	41 mg/dL	7 to 27	21	21
Phosphorus	7.3 mg/dL	2.5 to 6.8	6.4	5.4
Calcium	10.8 mg/dL	7.9 to 12.0	9.5	10.3
Sodium	147 mmol/L	144 to 160	149	152
Potassium	4.9 mmol/L	3.5 to 5.8	5.3	3.8
Chloride	109 mmol/L	109 to 122	109	111
Total protein	6.2 g/dL	5.2 to 8.2	5.7	6.4
Albumin	2.8 g/dL	2.2 to 3.9	3.0	3.0
Globulin	3.4 g/dL	2.5 to 4.5	2.7	3.4
ALT	93 U/L	10 to 125	169	226
ALP	334 U/L	23 to 212	1,193	1,177
GGT	5 U/L	0 to 11	19	16
Bilirubin	0.4 mg/dL	0.0 to 0.9	0.1	0.5
Cholesterol	215 mg/dL	110 to 320	321	294
Amylase	312 U/L	500 to 1,500	318	(not printed)
Lipase	439 U/L	200 to 1,800	423	(not printed)

The analyzer comment: increased RDW, anisocytosis present, review the blood film. Amylase is below the printed interval on 15 Aug.

Emergency assessment as written: increased respiratory rate and effort (bronchitis with bronchopneumonia versus pulmonary hypertension with lower-airway disease versus other); lethargy and legs giving out (intracranial versus cardiac/syncope versus other); murmur; history of HAC; history of DI.

What was done (names only): intravenous fluids, oxygen, an antiemetic, two antimicrobials, pimobendan, theophylline continued, NPO, respiratory watch, seizure watch. Neurology and cardiology referral to a tertiary hospital was offered and declined that night. Overnight hospitalization was accepted.

17 Aug 2023, internal medicine discharge: diagnosis list: arrhythmia; history of heart disease; history of chronic respiratory disease; history of suspected Cushing’s and DI. She improved the first night, then had several collapse episodes with a very fast supraventricular rhythm that slowed with treatment. Abdominal ultrasound: no splenic or adrenal mass. Priority

stated as the heart. The internist also wrote that disorientation plus the prior Cushing's and DI labels could relate to a pituitary tumor, and that arrhythmia itself can cause disorientation. Sent home on prednisolone, sotalol, pimobendan, and an antiemetic. Recheck 3 to 5 days. Cardiologist as soon as possible.

That evening: returned in distress. Euthanasia at the owner's request. Private cremation.

Pause and think

1. Write a problem list for this admission. Which problems are most urgent, and which could wait?
2. Collapse and disorientation could be cardiac, neurologic, or metabolic. What data separates them, and what would you do first?
3. Describe the leukogram. What three things can produce this pattern in Coco?
4. Erythrocytosis with a low MCV and a high RDW: give two explanations (hydration, hypoxemia, iron).
5. She was discharged on prednisolone, and an earlier bronchitis drug also contained a corticosteroid. What are the risks in a dog with a Cushing's label and an arrhythmia?
6. The owner declined referral at first. How would you discuss referral, cost, and prognosis, and how would you support a decision for euthanasia?
7. Looking back over the whole record, which three decisions or missing data points would you change?
8. The radiology service called her a bulldog. The emergency service called her a Beagle. What else in a STAT report do you now re-read?

Putting Coco together

Build a master problem list. For each problem, record the evidence, whether it is active, your leading differential, the next test, and what you would treat now.

Problem	Evidence	Active?	Leading differential	Next test	Treat now?
Polyuria and polydipsia	8 cups/day; USG 1.006 to 1.011				
Conflicting urine culture	Strep one day, staph the next; later "no bacteria"				

Problem	Evidence	Active?	Leading differential	Next test	Treat now?
Hyperadrenocorticism label	Face sheet; high ALP; trilostane then stopped				
Diabetes insipidus label	Face sheet; desmopressin trial with confounders				
Night waking / “dementia”	Apr 2023 label; nocturia; pain; drugs; heart/lungs				
Weight loss then rebound	30.4 to 23.8 to 29.6 lb				
Hepatopathy label	ALP 875 then >1,100 then 334				
Heart disease / arrhythmia	Grade IV murmur; VT note then SVT note				
Airway / pneumonia	Stridor; DACVR bronchitis and bronchopneumonia				
Mineral / vitamin D supplements	D3 and calcium lactate; phosphorus 7.3 at the end				
“Hyperthyroidism” label	Face sheet plus a thyroxine bottle				

Reasoning traps in this chart

Trap	Where it appears
Anchoring on old-dog dementia	Cognitive dysfunction entered while treatable contributors were active
Premature closure	Finding and treating a UTI before the polyuria was explained
Normal creatinine means normal kidneys	Creatinine 0.3 to 0.8 in a polyuric, thin, aging dog

Trap	Where it appears
Diagnostic momentum	Cushing's, DI, hepatic disease, and hyperthyroidism carried forward with no test on file
Confounding by simultaneous change	Three changes in two weeks in Jun 2022
Two cultures, one story	Hemolytic Streptococcus and Staphylococcus one day apart
Polypharmacy	Diuretic, corticosteroid, sedatives, bronchodilator, vitamin D, desmopressin
Reference intervals that change	Antech ALP 5 to 131 versus IDEXX ALP 23 to 212
Treating a number	Calcium and vitamin D with no PTH, ionized calcium, or diet analysis on the PDFs
Competing priorities	Heart, brain, and adrenal disease demanding different drugs
Banner versus vitals	14.1 kg versus 14.8 kg; Beagle versus bulldog on the radiology header
Lecture slide versus laboratory PDF	PTH and calcium figures on the slide that are not on the IDEXX/Antech pages

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The common thread

	Oliver	Coco
Behavioral phenotype	Hallway arousal, lobby freeze, concierge catecholamine surge	Restless, disoriented, vocal at night
Medical screen	Overview slide: later normal CBC, chemistry, thyroid. Short deck: parvo at 11 weeks, then cleared	Overlapping endocrine, urinary, cardiopulmonary, and mineral problems
Functional axes	Locus coeruleus, PAG, amygdala; sympathoadrenal	Endocrine, metabolic, urinary, cardiac, pulmonary
First priority	Localize the step; train from the hallway; do not skip a medical screen	Treat the medical drivers of night-time restlessness, then reassess cognition
Trap	Calling a lobby freeze "just drama," or calling 11-week parvo the whole diagnosis	Calling nocturia and panting "dementia" after a UTI is cleared

Bodaway (the Irish Wolfhound mix with refractory itch) remains in the original short pre-lecture if you were assigned that case. The same hierarchy applies: complete the pruritus workup, treat confirmed abnormalities, and do not call chronic scratch a habit.

Prepare for class

Oliver

1. One-sentence problem representation that includes hallway, lobby, and concierge.
2. Three findings that would make the case no longer situational only.
3. Which row of the pharmacology table matches the lobby freeze, and which row matches Coco’s dementia label.
4. The two sentences you would say to the owner to separate a fear response from a moral failure, and the two sentences about gold-standard parvo care versus the care you can deliver.

Coco

1. Write a one-sentence problem representation for Jan 2022, Jan 2023, and 15 Aug 2023.
2. Give the five most likely causes of her night-time restlessness in Jan 2023. Rank them. Name the single piece of data that would most change the ranking.
3. Explain what the normal renal values rule out and what they do not. Use the June 2022 UTI as your example.
4. List the observed findings, interpretations, causal hypotheses, and diagnostic conclusions about her mineral status that the chart supports. Mark the box for each.
5. Propose a stepwise plan for a polyuric, night-waking dog that sequences medical correction before behavioral and cognitive treatment.
6. Write the two or three sentences you would say to the owner to separate a working hypothesis from a confirmed diagnosis.
7. Reconcile the lecture-slide PTH/calcium figures with the IDEXX/Antech pages in this packet. Which box is each number?

Bring your written answers and one question about something in this reading that did not make sense.

Sources used to build this packet

- Integration Medicine III short Oliver deck (7 slides): building-sequence investigation, 4 Jan 2021 and 25 Jan 2021 parvo records, problem-list slide, pharmacology recap. Overview lecture “Am I a Drama?” for the later 2-year-old / elevator / normal-lab lines, marked as not reprinted on the short deck. Coco lecture PTH and calcium figures marked as unverified against the laboratory PDFs.
- Teaching figure (`IM3_Coco_Metabolic_Lock_Figure.png`): two-panel conventional-CDI path versus proposed secondary nDI / secondary HPT. Lecture-note PTH and USG values asterisked. V2 / AQP2 labeled as hypothesis, not as a receptor assay.
- De-identified primary-care chart extracts (Jan 2022 through Jun 2023), including the June 2022 cystocentesis/culture sequence and the desmopressin USG series.
- Antech Vet Screen and UPC (27 to 28 Jan 2022; 31 Mar 2022).
- Nelco allergy panel (14 to 15 Sep 2021), food list only as printed.
- IDEXX ProCyte Dx and Catalyst One (15 Aug 2023) with the analyzer comparison columns from 13 Jun 2023 and 22 Jun 2023.
- Emergency initial exam (15 Aug 2023), STAT radiology report (15 Aug 2023), internal-medicine discharge (17 Aug 2023), and euthanasia note (17 Aug 2023).

No milligram-per-kilogram in this packet is a standing order. Confirm every number in Plumb or the hospital protocol. Neither this reading nor a formulary replaces a veterinary license.